Investigating the limits and interactions of navigational systems in a multi-agent environment

George Stanway
(psygs6@nottingham.ac.uk)

Abstract

This study explores how individual and collective navigation systems found in nature allow multiple agents to efficiently search for and retrieve resources. The main focus was identifying the limitations both systems face, how they interact, and how to tune agent and environmental parameters to overcome them. To do this, I built a GPU-accelerated ant simulator using the Godot game engine to simulate and interact with thousands of agents in real time. Agents place pheromones to lead each other towards food or back home (stigmergy), but can individually navigate home using a continuously updated internal compass (odometry). I designed tests to test curiosity driven exploration, path optimisation, odometric noise resilience, and environmental adaptation. Quantitative results show that while individual odometric error statistically hinders colony efficiency, stigmergic communication makes up for this. Testing shows that path optimisation speed is proportional to the pheromone decay rate, with longer lasting pheromones allowing for faster convergence to the shortest path, whereas recovery speed has a non-linear inverse relationship, since longer lasting pheromones delay exploration.

Contents

1 Introduction **2**

2 Design and Implementation **3**

2.1 Environment 3

2.2 Agents 4

3 Testing and Results **6**

3.1 Test 1: Exploitation vs Exploration 6

3.2 Test 2: Path Optimisation (Shortest Path) 8

3.3 Test 3: Noise Resilience 9

3.4 Test 4: Dynamic Environment Adaptation 11

4 Discussion **12**

5 Conclusion **13**

1 Introduction

Autonomous intelligent agents are programs that operate in an environment, perceive the environment, and choose actions to accomplish tasks. The agents in this study are ants, each one a GPU thread in my Godot game engine simulator [5], with their behaviour determined by a state machine. They can detect nearby pheromones and food to make reactive decisions, and manage internal state (displacement from colony) to make proactive actions.

This project is a classic example of swarm intelligence, which is useful for solving many real world problems (such as finding a shortest path) using simple local rules. Stigmergy is the method of decentralised coordination the agents use [4], allowing them to work together without leaders by placing pheromones that others can follow. It forms the basis of Ant Colony Optimisation (ACO) algorithms [3]. This method acts as a vital global memory, but can be misleading or persist too long. Pheromone trails can form loops, and ants that only use stigmergy for navigation (such as the Army ant) can get stuck in infinite 'death spirals' [1]. Path integration is the method used by the agents (and the Saharan silver ant) for odometry [9], which allows them to track their displacement from the colony over time and escape these loops. However, this estimate is prone to error over time (known as 'cognitive drift' [2]) so agents relying on it alone will eventually become completely lost.

The interaction between stigmergy and odometry illustrates the stability-plasticity dilemma. A successful swarm must retain enough memory to exploit known resources (stability) while remaining flexible enough to abandon obsolete trails and adapt to sudden environmental changes (plasticity) [6]. I encountered all of these real-world computational challenges whilst building the simulator, and formulated four research questions to investigate the limits of these interacting navigational systems:

Question 1: To what extent does the trade off between curiosity driven exploration and pheromone driven exploitation affect a colony's efficiency at collecting food in a resource scarce environment? This question aims to find the optimal rate of ignoring pheromone trails (curiosity) to maximise resource collection.

Question 2: How does the pheromone decay rate influence the effectiveness of stigmergy? This question investigates the minimum pheromone persistence necessary for stability (remembering useful trails) before agent behaviour degrades to random wandering.

Question 3: How detrimental is individual odometric error in the efficiency of a colony at collecting food, and how effective is stigmergic communication at avoiding this? This question aims to test whether pheromone trails are effective enough to overcome odometric error, and quantify how important stigmergy is.

Question 4: What is the relationship between the persistence of pheromones and the adaptiveness of a colony when subjected to sudden environmental changes? This question investigates the maximum pheromone persistence possible for plasticity (forgetting harmful trails) before agents are distracted by dead ends.

2 Design and Implementation

2.1 Environment

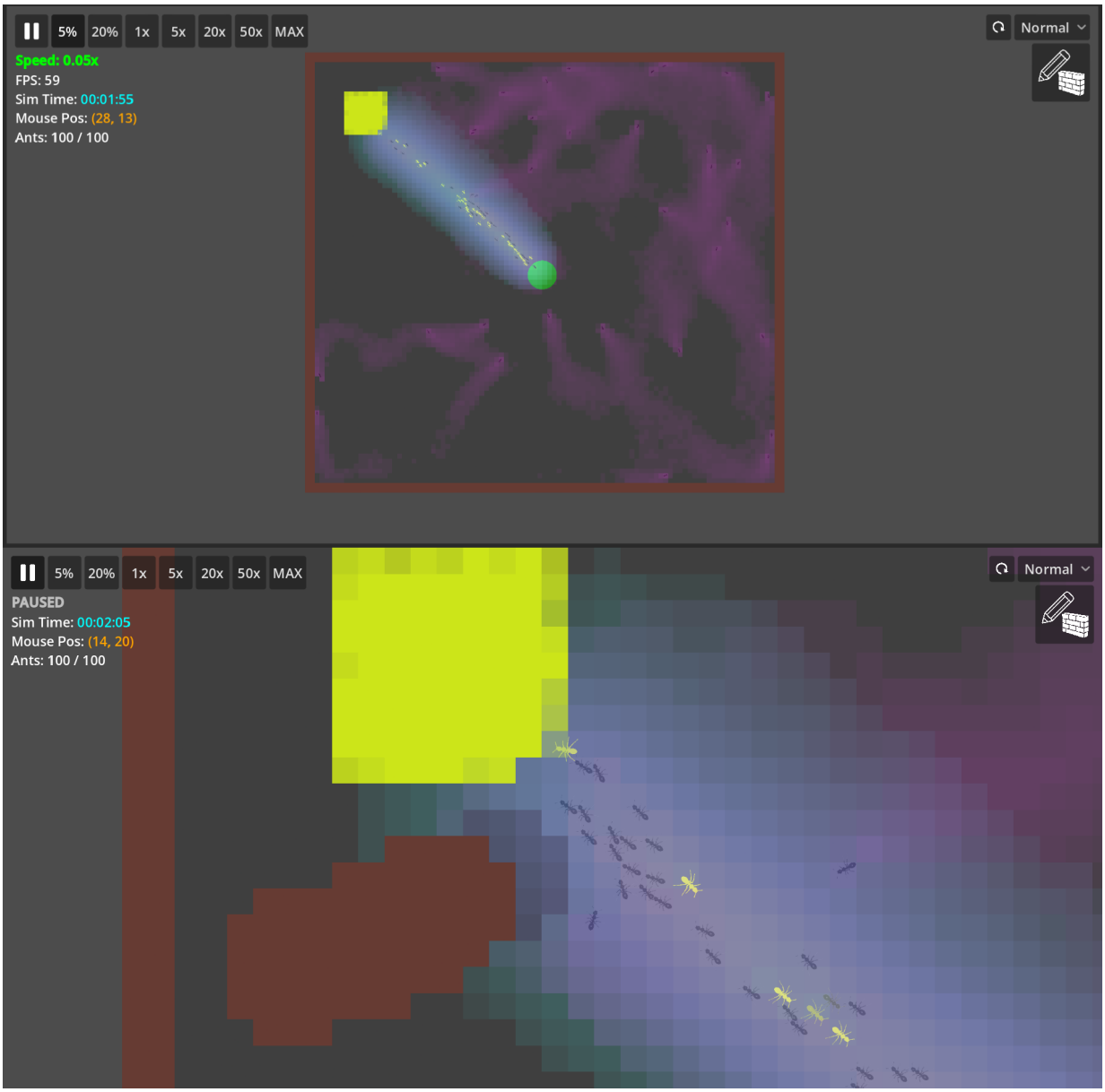


Figure 1: Screenshots of my custom simulator running

The screenshots above show the simulator running. The top image shows a zoomed out view of one of the maps, with a highway already formed from the colony (green circle) to the food (yellow square). The blue pheromones are placed by (yellow) ants carrying food, and lead (black) searching ants to food. The purple pheromones are placed by searching ants, and lead ants with food back home.

The bottom image shows a zoomed in view of the same map demonstrating the dynamic editing abilities of the simulator. The user is midway through painting a wall to block off the path to the food, but could also draw food. Time can be controlled with the top-left buttons, reaching 200 times speed with 60 fps at max, and the current map can be reset or changed with the top right buttons.

The environment is a single image, represented as a Godot TileMapLayer on the CPU and sent to the GPU, with each pixel representing a single cell of the map and each channel (RGBA) storing information about that cell. Red stores walls (either 0.0 or 1.0), Green stores the food value, Blue stores the 'to food' pheromone value, Alpha stores the 'to home' pheromone value.

The simulator uses 3 main compute shaders (and one for painting) to function. The first one (ants_build.glsl) rebuilds a spatial hash grid containing all the ants, allowing for fast lookups of ants close by. The next shader (ants_process.glsl) runs for every ant and processes them, I'll explain in the next section. The last shader (map_process.glsl) runs for every cell in the world and applies the reaction-diffusion equation **[8]** as a 3x3 convolution pass to evaporate and diffuse pheromones. Since pheromone values are stored as unsigned integers, stochastic rounding is used to ensure mass is conserved over time.

2.2 Agents

Agent logic is processed each tick by the main GPU compute pass (ants_process.glsl). This pass handles physical kinematics: reading from the spatial hash map to find nearby agents, running ant-to-ant separation **[7]**, resolving wall collisions and moving the agent. The behaviour function is then run (pseudocode below) which handles agent state transitioning and rotation in response to stimuli (like turning to food or to follow gradients). Cognitive drift is then applied via Gaussian noise to the agent's polar odometry. Finally, the agent safely drops pheromones to the world texture using 'atomicAdd'.

Algorithm 1 Agent Behaviour and Hybrid Steering Logic

```
procedure UpdateBrain(ant, idx)
    if IsStuck(ant) then
        TurnAround(ant)                                  ▷ Turn around if not moving
    end if
    // Resource interactions
    if ant.state == GATHERING then
        TakeFoodAtomic(idx, ant)                         ▷ Thread-safe Compare And Swap
        if IsFull(ant) then
            ant.state ← RETURNING                        ▷ Full, go home
        end if
        return
    else if ant.state == DROPPING then
        RecalibratePedometer(ant)                        ▷ Zero-out path integration error
        DropFoodAtomic(idx, ant)
        if IsEmpty(ant) then
            ant.state ← SEARCHING                        ▷ Empty, find food
        end if
        return
    end if
    // Task transitions and avoidance
    if FoundFood(ant, idx) then
        ant.state ← GATHERING                            ▷ Start picking up food
        return
    end if
    if ReachedColony(ant) then
        ant.state ← DROPPING                             ▷ Start dropping off food
        return
    end if
    if FacingWall(ant) then
        SteerAwayFromWall(ant)                           ▷ Reactive obstacle avoidance
        return
    end if
    // Hybrid navigation (Stigmergy vs Odometry)
    steered ← TryFollowPheromones(ant)                   ▷ Try to follow pheromones/food
    towards_colony ← IsFacingColony(ant)                 ▷ Check internal compass vector
    if ant.state == SEARCHING then
        misled ← steered ∧ towards_colony                ▷ Following trail towards colony?!
        if misled ∧ FarFromColony(ant) then
            BreakDeathSpiral(ant)                        ▷ Reject death loops / residual trails
        else
            ApplyWanderJitter(ant)                       ▷ Curiosity-driven exploration
        end if
    else                                                 ▷ State: RETURNING
        misled ← steered ∧ ¬towards_colony               ▷ Following trail away from colony?!
        if misled then
            BreakDeathSpiral(ant)                        ▷ Override stigmergy with odometry
        end if
        weight ← steered ? 0.05 : 0.01                   ▷ Trust pheromones more if present
        ApplyOdometricPull(ant, weight)                  ▷ Steer towards internal home vector
    end if
end procedure
```

3 Testing and Results

To ensure the validity and reproducibility of the test results, these default settings were established beforehand and maintained across all simulations to act as a control. Only the settings starting with '*' were varied during testing, and only one per test:

Table 1: Global Default Settings

Parameter	Value	Description
Environment		
Max pheromone / food	1000	Saturation limit per grid cell
*Decay rate	0.1	Evaporation rate of pheromones
Diffusion rate	0.02	Spatial spreading of pheromones
Spatial grid cell size	2.0	Resolution of the spatial hash grid
Agent Traits		
Ant count	100	Colony population
Colony radius	4.0	Radius in grid cells
Base speed	2.0	Movement speed (units per second)
Pheromone drop rate	100 / sec	Pheromones dropped per second
Recruitment multiplier	2.0	Intensity multiplier for to-food pheromones
*Odometric error	0.05 / sec	Cumulative path-integration noise
*Curiosity	0.1	Probability of ignoring pheromone trails
Max carry capacity	50	Maximum food carried per agent
Interaction time	1.0 s	Duration to pick up or drop food
Simulation		
Tick rate	60 TPS	Engine ticks per second ($\Delta t = 1/60s$)
Metrics interval	5.0 s	Frequency of telemetry snapshotting

To address the four research questions, four distinct tests were designed. For each test, an independent variable was swept across a defined range over multiple batches to ensure statistical significance.

3.1 Test 1: Exploitation vs Exploration

This experiment evaluated the optimal balance between collective exploitation (following pheromones) and individual exploration. The independent variable was *curiosity*. The map was very large with a single small patch of food far from the colony, to penalise inefficient searching and retrieving.

Parameter	Configuration
Independent variable	Curiosity ($0.0 \rightarrow 1.0$)
Dependent variable(s)	Ticks to first food, Total food collected (per sweep)
Sweeps $\times$ Batches	40×40 (1,600 total simulations)
Run time	8.5 minutes (510 seconds)
Special conditions	None

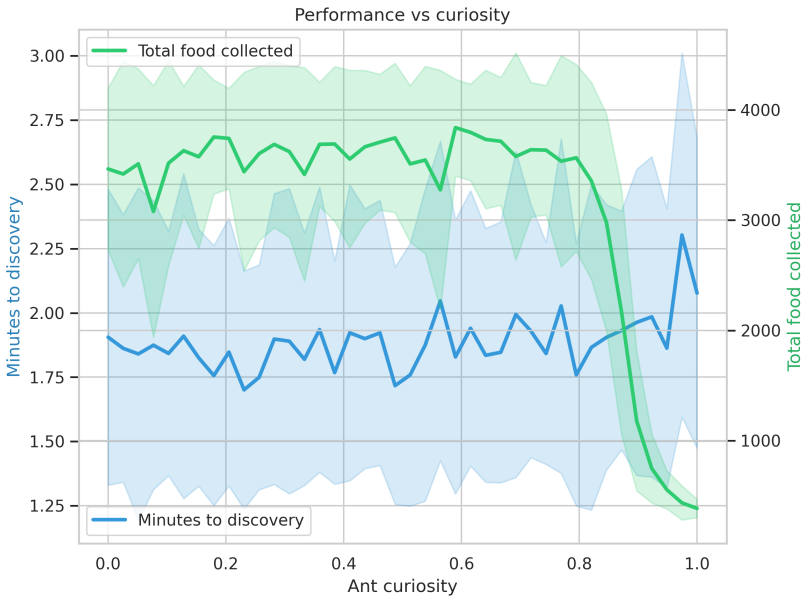


Figure 2: Comparison of total food collected and time to initial discovery across varying curiosity rates.

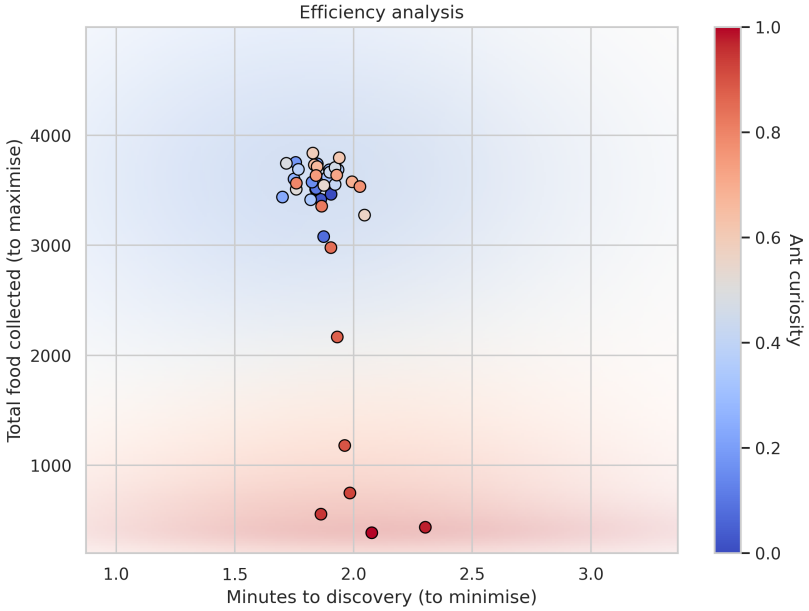


Figure 3: Scatter plot mapping the efficiency of each curiosity configuration. The background colour maps to average curiosity and opacity reflects the standard deviation of the batches.

Evaluation:
The results demonstrate a clear threshold effect regarding agent curiosity. As seen in Figure 2, total food collection remains remarkably stable and high for curiosity values between 0.0 and 0.75. The time to initial discovery (blue line) also remains relatively flat, indicating that high curiosity is not strictly necessary to locate sparse resources.

However, a phase transition occurs beyond $C = 0.8$. At this threshold, the exploitation of pheromone trails collapses. Figure 3 confirms that while low curiosity ($C < 0.4$, represented by blue/white dots) consistently yields high efficiency, hyper-exploration ($C > 0.8$, dark red dots) prevents the swarm from following established highways, resulting in colony starvation. In this specific sparse environment, strict collective exploitation significantly outperforms high individual exploration.

3.2 Test 2: Path Optimisation (Shortest Path)

This test investigated the relationship between pheromone persistence and the emergence of shortest-path convergence. The independent variable was the *decay rate*. To measure optimisation, the environment featured shorter and longer paths to the same singular food source.

Parameter	**Configuration**
Independent variable	Pheromone decay rate ($0.01 \rightarrow 0.8$)
Dependent variable(s)	Short path traffic, Long path traffic (per frame)
Sweeps $\times$ Batches	30×50 (1,500 total simulations)
Run time	8.5 minutes (510 seconds)
Special conditions	None

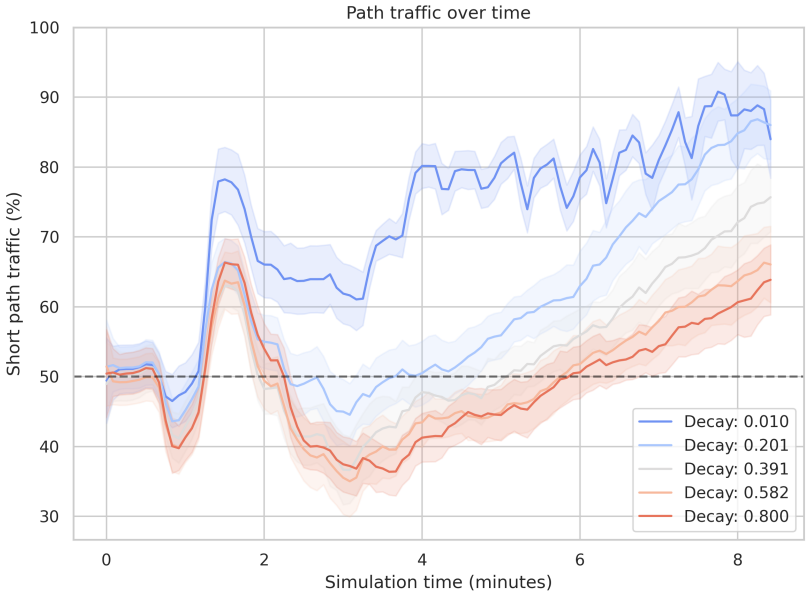


Figure 4: Time-series data showing the percentage of ants choosing the shorter path. Lower decay rates allow for faster convergence.

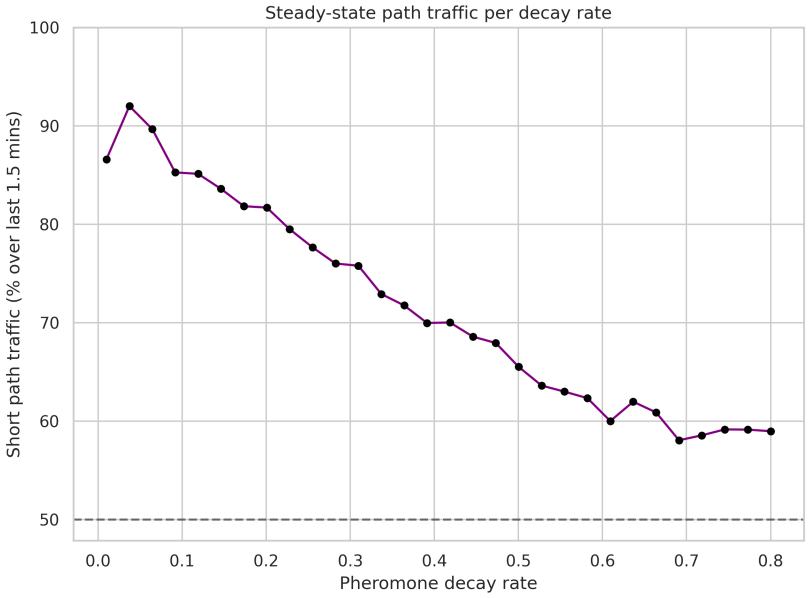


Figure 5: Steady-state short path traffic averaged over the final 1.5 minutes of the simulation, plotted against the pheromone decay rate.

Evaluation:
Test 2 highlights the vital role of pheromone persistence in path optimisation. Figure 4 illustrates that colonies with highly persistent pheromones (decay rate $d = 0.010$) successfully converge on the shortest path, reaching up to 90% of traffic. Conversely, rapidly evaporating trails ($d = 0.800$) struggle to differentiate between the two paths, failing to exceed a 60% distribution.

Figure 5 isolates this steady-state traffic, revealing a strong inverse correlation between the decay rate and path optimisation. Longer-lasting trails provide the necessary memory for the colony to iteratively optimise its routing, as the shorter path naturally accumulates a higher density of pheromones per second.

3.3 Test 3: Noise Resilience

This test evaluated how effectively stigmergy makes up for individual cognitive drift. The *odometric error* was linearly increased. Importantly, this test was run in two versions (with and without pheromones enabled) to isolate their importance. The map was a regular size with food scattered around.

Parameter	**Configuration**
Independent variable	Odometric error ($0.0 \rightarrow 0.8$)
Dependent variable(s)	Total food collected (per sweep)
Sweeps $\times$ Batches	40×40 (1,600 total simulations per version)
Run time	3.5 minutes (210 seconds)
Special conditions	**2 versions:** Pheromone drop rate = 0 vs 100

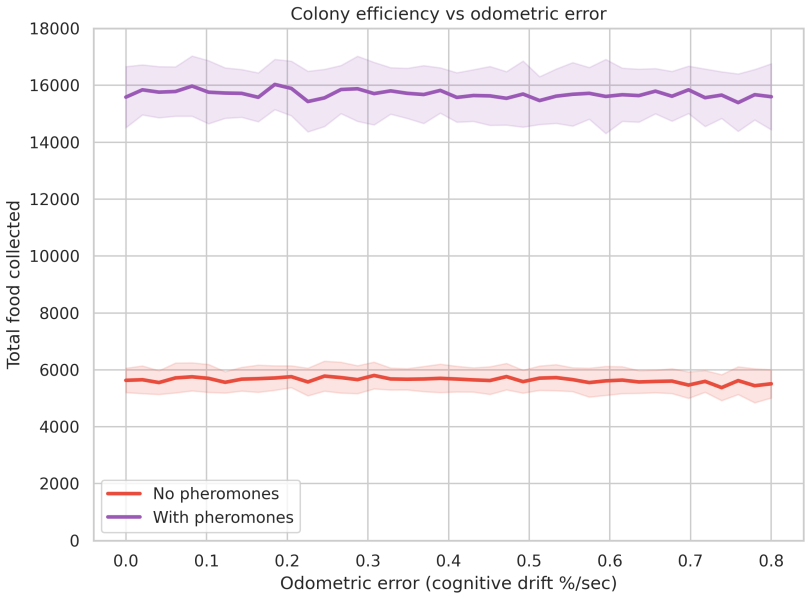


Figure 6: Colony efficiency compared between a purely odometric swarm (red) and a hybrid stigmergic swarm (purple) under increasing cognitive drift.

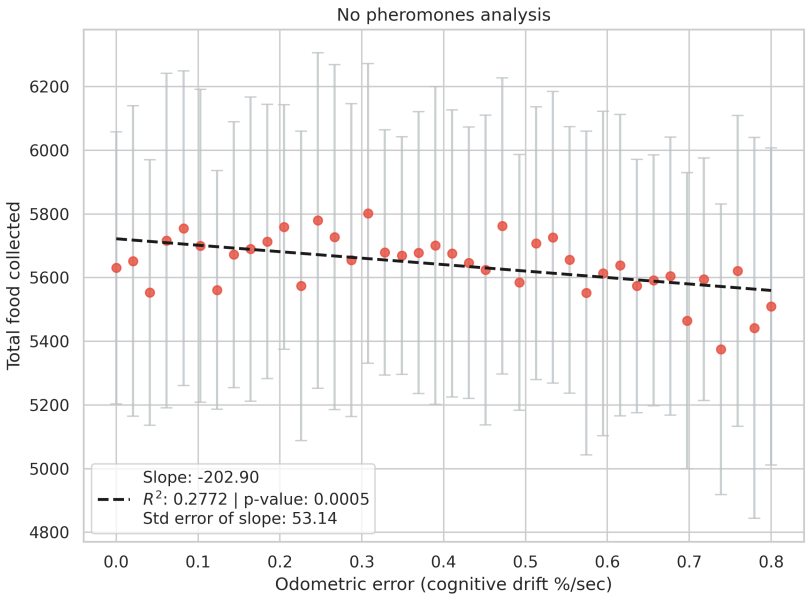


Figure 7: Linear regression analysis of the control group (no pheromones), showing a statistically significant decline.

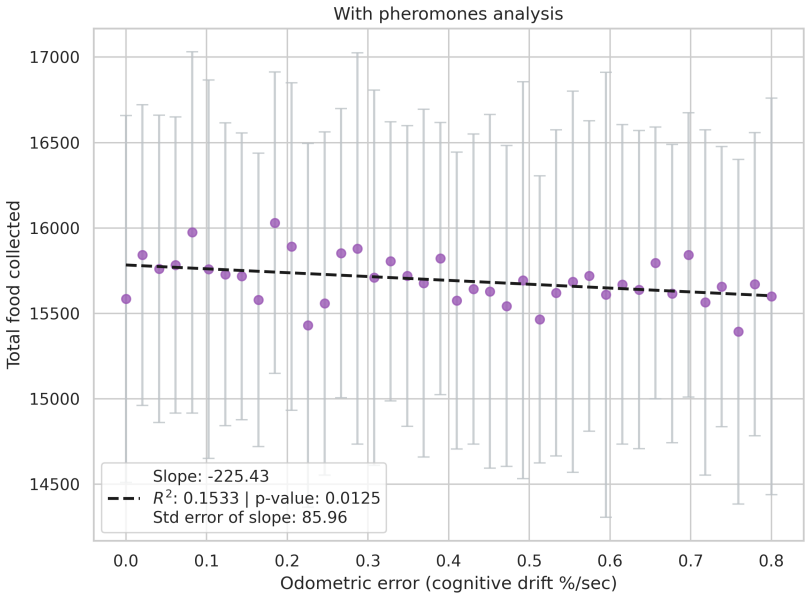


Figure 8: Linear regression analysis of the experimental group (with pheromones), showing the stigmergic buffering effect.

Evaluation:
This test quantifies the buffering capacity of stigmergy against individual cognitive drift. Figure 6 shows a large performance difference. The colony using pheromones was almost 3 times more efficient than those relying solely on path integration.

Figures 7 and 8 break down the linear regressions. Without pheromones, odometric error causes a statistically significant decline in foraging efficiency ($R^2 = 0.277, p = 0.0005$). While the stigmergic colony also exhibits a negative slope as individual error increases, the performance boost completely outweighs the individual odometric losses. This is good evidence that stigmergic communication acts as an effective noise-mitigation strategy, overriding faulty internal compasses with reliable global data.

3.4 Test 4: Dynamic Environment Adaptation

A closer larger food source is deleted after 5 minutes (allowing the colony to form a pheromone highway), forcing them to adapt and start retrieving the second food source further away. The *decay rate* was swept to determine the optimal pheromone persistence (stability) required to successfully abandon a dead trail and recover quickly (plasticity).

Parameter	**Configuration**
Independent variable	Pheromone decay rate ($0.01 \rightarrow 0.8$)
Dependent variable(s)	Active foraging ants (per frame)
Sweeps $\times$ Batches	20×100 (2,000 total simulations)
Run time	12.0 minutes (720 seconds)
Special conditions	**Dynamic event:** Food source at (60, 30) deleted mid-run

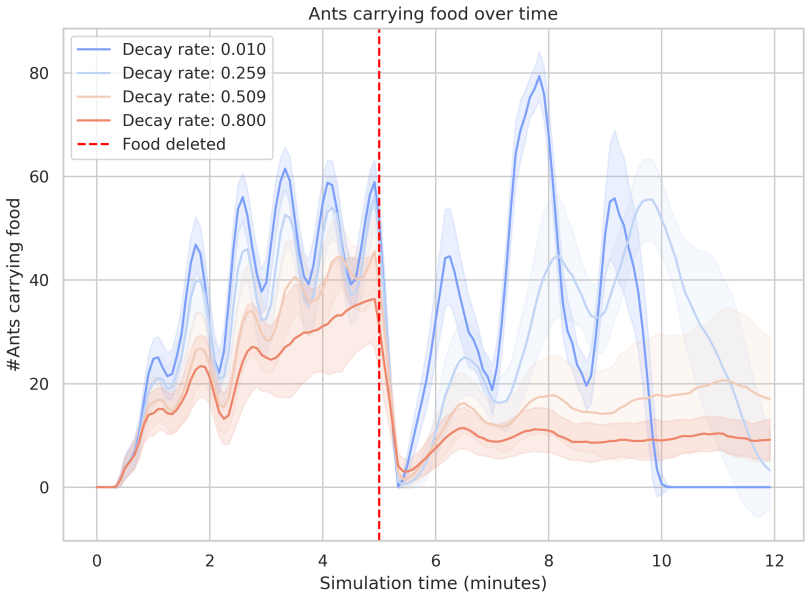


Figure 9: The number of ants carrying food (active foragers) over time. The vertical dashed line marks the sudden deletion of the primary food source, forcing the colony to adapt.

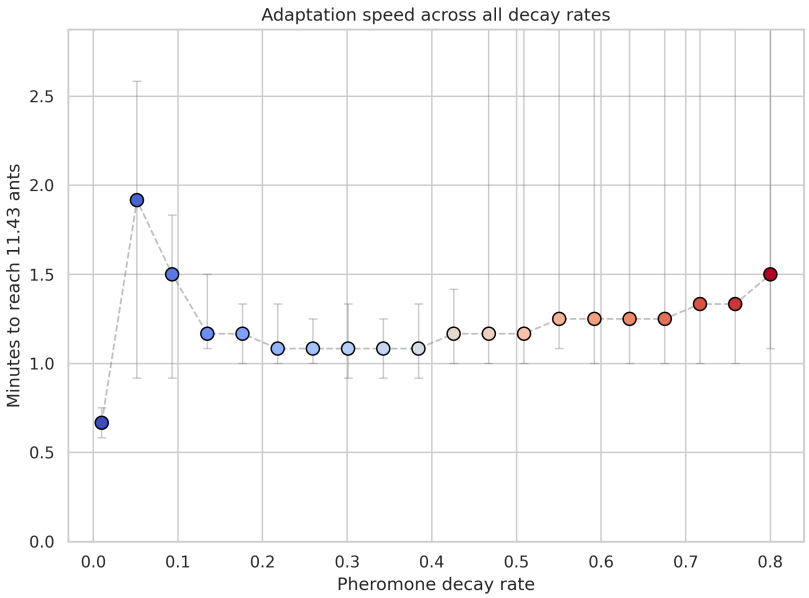


Figure 10: Adaptation speed (time taken to recover a baseline foraging rate) across different decay rates, illustrating a complex non-linear relationship.

Evaluation:
Test 4 demonstrates the stability-plasticity dilemma. After the primary food source is deleted at $t = 5$ minutes (Figure 9), a very high or low persistence of pheromones (stability) harms the speed at which the colony recovers (plasticity).

Figure 10 reveals a distinctly non-linear relationship regarding the colony's adaptation speed. A near-zero decay rate recovered almost instantly, due to trails from ants who happened to find the further food from the start lasting the full 5 minutes, and ants hopping over to this immediately (stability so high that plasticity is irrelevant). A low decay rate ($d \approx 0.05$) takes the longest to recover, due to the highway between the closer food lingering after its deletion and directing ants back to the colony (stability harming plasticity). Then as decay rate increases ($d \approx 0.2$ to 0.4), recovery time decreases non-linearly to an equilibrium minimum of about 1.2 minutes. It is here that stability is balanced with plasticity. Increasing decay rates further ($d > 0.5$) decreases stability enough to hinder highway establishment and cause recovery time to increase again, since ants start getting lost on their way to the further food.

4 Discussion

The results of the testing provide clear quantitative answers to the four central research questions regarding hybrid navigational systems.

Question 1: The data heavily favours collective exploitation in resource-scarce environments. High individual curiosity ($C > 0.8$) completely disrupted the formation of stigmergic highways, causing the colony's foraging efficiency to collapse. For a swarm to be efficient, agents cannot fully ignore pheromone trails.

Question 2: The speed and success of path optimisation are directly proportional to pheromone persistence. Highly stable pheromones (low decay) provided the necessary memory for the swarm to iteratively recognise and commit to the shorter route. If the decay rate was too high, the system took far longer to converge to the more efficient path.

Question 3: Individual odometric error was shown to cause a statistically significant decline in performance. However stigmergic communication acted as an adequate buffer against this cognitive drift. By allowing agents to place and follow pheromones, the swarm successfully corrected the navigational errors of individual agents, causing a nearly 300% increase in efficiency compared to a purely odometric system.

Question 4: The simulation clearly demonstrated the stability-plasticity dilemma. While high stability

(low decay) was necessary for path optimisation (Q2), test 4 showed that it severely hindered the colony's ability to adapt (plasticity) when the environment changed. Oppositely, high plasticity (high decay) allowed agents to quickly forget dead trails but prevented them from forming new, stable highways. The swarm achieved optimal adaptability only within a narrow, moderate decay range ($d \approx 0.2 \rightarrow 0.4$), balancing the retention of useful memory with the flexibility to discard obsolete data.

5 Conclusion

This study successfully investigated the limits and interactions of individual odometry and collective stigmergy using a custom, high-performance GPU simulator. The results demonstrate that neither system is flawless in isolation, path integration suffers from cumulative cognitive drift and purely stigmergic navigation is vulnerable to infinite looping. However, when combined into a hybrid navigational model, the systems compensate for each other's weaknesses. The collective memory buffers the errors of the individual, while the individual's internal compass provides the necessary baseline to break free from stigmergic death spirals. The efficiency of this hybrid system relies entirely on tuning the pheromone decay to match the volatility of the environment, successfully navigating the stability-plasticity dilemma.

AI Statement

Generative AI tools were used for LaTeX formatting and proofreading. All raw data, graphs, simulator and shader code were (painstakingly) made by me.

References

[1] Daniel Angus and Colin Woodward. Balancing between exploration and exploitation in aco. In *Proceedings of the IEEE Congress on Evolutionary Computation*, 2009.

[2] S. Deeti and K. Cheng. Desert ants (melophorus bagoti) oscillate and scan more in navigation when the visual scene changes. *Animal Cognition*, 2025.

[3] Marco Dorigo and Thomas Stützle. *Ant Colony Optimization*. MIT Press, 2004.

[4] Pierre-Paul Grassé. La reconstruction du nid et les coordinations inter-individuelles chez Bellicositermes natalensis et Cubitermes sp. la théorie de la stigmergie. *Insectes Sociaux*, 6(1):41–80, 1959.

[5] Juan Linietsky, Ariel Manzur, and Godot Engine Contributors. Godot engine. https://godotengine.org, 2014. Source code available at https://github.com/godotengine/godot.

[6] Michalis Mavrovouniotis, Felipe M. Müller, and Shengxiang Yang. Ant colony optimization with self-adaptive evaporation rate in dynamic environments. *2014 IEEE Symposium on Swarm Intelligence*, pages 1–8, 2014.

[7] Craig W Reynolds. Flocks, herds and schools: A distributed behavioral model. *ACM SIGGRAPH computer graphics*, 21(4):25–34, 1987.

[8] Alan Mathison Turing. The chemical basis of morphogenesis. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, 237(641):37–72, 1952.

[9] Rüdiger Wehner. Desert ant navigation: how miniature brains solve complex tasks. *Journal of Comparative Physiology A*, 189(8):579–588, 2003.